

Chapter 14

Posture verbs in locative and existential predication across three Australian languages

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In this chapter, we examine the role of posture verbs in locative and existential predication in three Australian Aboriginal languages: Kukatja (Western Desert, Pama-Nyungan), Kune (a variety of Bininj Kunwok; Gunwinyguan), and Wik-Mungkan (Paman, Pama-Nyungan). We detail the semantic range of each posture verb and consider the extent to which individual posture verbs are recruited in more grammaticalised roles. Posture verbs exist alongside verbless predication strategies in all three languages and the frequency of their use differs across the three languages in notable ways depending on the animacy of the figure referent. We utilise standardised elicitation tasks to quantify observed variation in posture verb use and find that in each language, different posture verbs show differing degrees of grammaticalisation.

1 Introduction

1.1 Background and aims

The basic human stances of sit, lie, and stand are likely encoded in all languages and are sometimes referred to as ‘primary postures’ (Newman 2022). Morphemes encoding these primary postures have been investigated cross-linguistically revealing recurrent patterns that further motivate their typological grouping beyond their shared function depicting universally shared human postures (see



Ameka & Levinson 2007: 850; Newman 2002a; Newman 2022). Typical shared features of this semantic class include: high frequency (with respect to other posturals), semantic extensions reflecting experiential realities (e.g. ‘lie’ extending to ‘sleep’) and certain differential paradigmatic features. One particular area of typological interest is the extent to which posture verbs are allowed – or are, in fact, required – when encoding the location or existence of a figure: functions often referred to as locative (x IS AT y) and existential (x EXISTS [AT y]).¹ Such uses can be exemplified in English by the posture verb *lie*, which can be used for locative functions (*the city lies to the east*) and existential functions (*a river lies to the east*). In English, such strategies are not the only construction type for predicating location and existence but rather constitute minor strategies compared to the more prevalent use of the copula *be*. The usage of posture verbs is thus rather specialised in English, tending to occur for only certain types of spatial scenes involving certain types of figures and grounds.

For a subset of the world’s languages, however, posture verbs are relied upon as an essential predication element within a “Basic Locative Construction” (Ameka & Levinson 2007). A language’s Basic Locative Construction is defined in accordance with a functional frame, namely, it is the unmarked construction type used to answer the question: “where is x ?”. In languages like Arrernte [aran1263], Dutch [dutc1256] and Goemai [goem1240], to express an unmarked predication notion BE AT, a posture verb must be used. Crucially, there are still very few detailed studies of individual Australian languages exploring how obligatory posture verbs are in languages that make use of them in this way, what motivates their use (e.g. animacy/geometric properties of figure) and precisely what their semantic contribution to various clause types is. Posture verbs in many languages have been shown to take on a range of additional functions in locative or existential contexts, such as: signifying temporal meanings other than present tense (Austin 1981: 104; Hercus 1994); marking aspect (Blansitt 1975: 25; Bybee et al. 1994; Kuteva 1999; Newman 2002b); marking emphasis or contrast (Nordlinger 1998: 179); categorising a located subject entity (Newman 2022: 7); disambiguating polysemous nouns (Wilkins 1989, 2006); conceptualising a located entity as a mass or individuated entity (Kaufman 2013; Newman 2022); contrasting temporary and permanent location (Haspelmath 2025); or signalling containment (Lemmens 2002). What all of these studies show is that, even if a posture verb may appear to be a “semantically neutral” element with respect to

¹The terms *figure* and *ground* were popularised by Talmy (1983) and in simple terms can be understood as the relation being located and the relation against which something is located respectively.

its postural specification in locative or existential constructions, such an element may still have separate, or additional, meaningful functions. Such findings suggest that posture verbs can vary considerably cross-linguistically, and motivate further detailed studies, especially for lesser described languages.

In this paper, we explore the role of posture verbs in locative and existential predication in three Australian Aboriginal languages. The existing literature for Australian languages provides two fairly well-known starting observations: i) that posture verbs are both diachronically and synchronically involved in copula constructions (Dixon 2002b); and ii) that locative and existential meanings are often encoded by the same construction type (Dixon 2002a: 241–242; Goddard & Harkins 2002: 224–229). Our aim is to extend the description of postural semantics among Australian languages, enriching earlier observations about the typological profile of the region. We present findings on three languages with which we are familiar: two Pama-Nyungan languages (Kukatja [kuka1246] and Wik-Mungkan [wikm1247]), and one Gunwinyguan language (Kune [gune1238]). Using novel data and existing descriptions for each of the surveyed Australian languages, we ask:

- What is the lexical semantic range of posture verbs? (§3.1)
- How are posture verbs recruited in the predication of location and existence? (§3.2)
- Are locative and existential predicates distinguished in the languages? (§3.3)

These descriptive questions give rise to the observation that posture verbs exhibit substantial semantic flexibility in the study languages, more than is suggested in many of the existing descriptions. We subsequently analyse the frequency of posture verb use in locative clauses (§3.4) in order to provide some insights into their versatility. To better orient the reader with respect to how posture verbs feature in locative and existential functions, we first review the previous literature on posture verbs in Australian languages and report frequently attested meanings, functions, and semantic extensions (§1.2). In (§2) we outline the methodology for the current study, introduce the three languages (Kukatja, Kune and Wik-Mungkan) as well as the data collection process. A terminological note of order before proceeding is that we initially refer to constructions that semantically predicate the closely related notions of location or existence simply as *locative-existential* although the status of these terms is considered more explicitly in (§3.3).

1.2 Posture verbs in Australian languages

As part of a survey of copula predication strategies in Australian languages, Dixon (2002b) notes a recurrent pattern of copula verbs having grammaticalised from erstwhile posture verbs in several independent contexts, while also drawing attention to the synchronic copula-like functions exhibited by posture verbs in many languages. Research specifically dedicated to posture verbs has also been conducted previously for the languages Arrernte [aran1263] (Goddard & Harkins 2002; Wilkins 1989, 2006), the closely related Western Desert languages Pitjantjatjara [pitj1243] and Yankunytjatjara [yank1247] (Goddard & Harkins 2002), Jaminjung [jami1236] (Schultze-Berndt 2006), Ngan'gityemerri [nang1259] (Reid 2002), and Warrwa [warr1258] (McGregor 2006).² Here we summarise the key parameters of observed variation in posture verbs from these and other works, in terms of: their degree of grammaticalisation, their role in locative and existential predication while also drawing attention to the additional roles they may serve in such contexts.

Aligning with global trends, the most common posture verbs are those encoding the so-called primary postures of 'sit', 'stand', and 'lie', although one does find languages which have additional lexical verbs that are treated as a primary posture on the basis of their aligned semantics and/or morphosyntax: for example *pupa-* 'be crouched over, bent over' in Pitjantjatjara and Yankunytjatjara (Goddard & Harkins 2002: 214); *yoongke* 'hang' in Kuuk Thaayorre (Pama-Nyungan, [thay1249]; Gaby 2017: 383) and 'perch' in Ngan'gityemerri (Reid 2002: 243); among others. Dixon's survey of copula constructions in Australian languages identified that posture verbs constitute the main stock of copula elements across the continent, with a smaller number of languages making use of a dedicated 'remain/reside' or 'go' verb in copula function (Dixon 2002b). While the use of posture verbs in copula predication varies across a range of parameters (e.g. functions, frequency, preferred posture verb, etc.), they are frequently core components of many locative and existential clauses (Dixon 2002b: 28).

In some Australian languages, the postural semantics of individual posture verbs show evidence of complete or near-complete bleaching when used in certain constructions. Consider examples (1) and (2) from central Australian languages Arabana and Warlmanpa respectively, where the posture verbs *thangk* and *ka* are used in identificational and presentational clauses respectively without any residual postural semantics evident (see also discussion in Dixon 2002b: 22).

²Arrernte (Arrandic), Pitjantjatjara (Western Desert), and Yankunytjatjara (Western Desert) are all Pama-Nyungan languages, the same language family to which Wik-Mungkan and Kukatja belong. The third language in this study, Kune, is unrelated to any of the aforementioned languages.

- (1) Arabana (Pama-Nyungan, [arab1267]; Hercus 1994: 295)
Antha minpaRu thangk-ka.
 1SG doctor **be(lit.sit)-PST**
 ‘I am a doctor.’
- (2) Warlmanpa (Pama-Nyungan, [warl1255]; Browne 2021: 385)
Yumpa=ju ka-nya kirtana.
 this=1SG.NS **be(lit.sit)-PRS** father
 ‘This is my father.’

The examples in (1) and (2) are notable for the unique argument structures that they instantiate in their respective languages; they are bivalent with two formally unmarked nominal arguments (i.e. they are in the non-overt absolutive case). For these languages, the availability of the [ABS-ABS] case frame is diagnostic of a posture verb having a copula use or a copula sense since such a case frame is unique to these verbal predicates. An interesting feature of posture verb systems in Australian languages is that typically just one of the posture verbs is recruited in bona fide copula roles (beyond locative-existential predication) of the kind illustrated in (1) and (2). Dench (1987: 210) makes this point explicitly for Martuthunira (Pama-Nyungan, [mart1255]), a language which permits all three of its posture verbs (*nyina*- ‘sit’, *karri*- ‘stand’, *wanti*- ‘lie’) to be used in copula function, but in which:

only *nyina*- can be said to function like a true dummy copula; both *karri*- and *wanti*- retain something of their core meaning in any copula construction [emph. our own].

Moreover, when one compares languages, there is substantial variation with respect to which member of this class has developed specialized functions (Dixon 2002a: 241–242): for some languages it is the ‘sit’ verb (e.g. Martuthunira: Dench 1987: 210 and Kunbarlang (Gunwiniguan, [kunb1251]): Kapitonov 2021: 307), in others it is ‘lie’ (as in Gooniyandi (Bunaban, [goon1238]): McGregor 1990: 310–311) or ‘stand’ (as in Pitjantjatjara and Yankunytjatjara: Goddard & Harkins 2002: 224). Dixon (2002b: 28) was the first to note that even amongst geographically and genetically close neighbours, individual languages appear to have developed these grammaticalised functions in a somewhat independent manner. In Pitjantjatjara and Yankunytjatjara (both members of the Western Desert subgroup of the Pama-Nyungan family), it is the ‘stand’ verb (both sharing a form *ngara*-) that has the broadest range of copula functions (Goddard & Harkins 2002: 224)

while in Kukatja, a northern member of the same tight-knit subgroup, it is the ‘sit’ verb *nyina-* (and not *ngara-* ‘stand’) that has the broadest range of functions. One finds very similar diachronic evidence from Maric languages in Queensland: Biri [biri1256] and Gidabal [cond1242] have a copula *wara-* which is cognate with ‘stand’ in neighbouring languages while Bidjara [bidy1243] and Gunja [guny1241] have a general copula *wiyi*, cognate with a ‘sit’ verb in neighbouring languages (e.g. Ngiyambaa [wang1291], Dixon 2002b: 28).

Australian Aboriginal languages also differ with respect to whether they require a posture verb in locative or existential clauses across the board or may freely omit the verbal component in favour of a truly verbless (viz. copula-free) construction. In Warrwa, for example, both the posture verb and figure NP may be omitted in locative constructions, as long as the ground nominal is retained (McGregor 2006: 123–124). On the other hand, a posture verb may be required in copula constructions but only in specific contexts. Posture verbs frequently operate as a semantically neutral “host” or “dummy” in order to encode verbal categories such as tense, aspect, and/or mood (Dixon 2002b: 17). For example, Wilkins describes the variable presence of posture verbs in copula constructions in Arrernte as being tense-dependent, posture verbs being required when non-present meanings are expressed (Wilkins 2006: 32). Similar descriptions of posture verb use in copula function have been made for Martuthunira (Dench 1987: 209–210) and Gooniyandi (McGregor 1990: 308–312). Differences in predication strategy have also been observed to be dependent on the nature of the locative component. In Ngiyambaa, Donaldson (1980: 232–233) reports that if the locative component involves a noun as a head, then a posture verb is required. If the locative component is a demonstrative or an interrogative, then a (lexically distinct) copula is used. It is unknown to what extent this pattern is unique to Ngiyambaa or has simply gone unnoticed due to incomplete documentation.

A further point of variation across the continent relates to how posture verbs in different languages show differing degrees of sensitivity to the geometric properties of inanimate figures. In Ngan’gityemerri, for instance, an inanimate figure that is “longer” in its horizontal dimension than its vertical dimension is said to ‘lie’ (e.g. rivers, roads or rocks) whereas other inanimate figures that are “taller” in their canonical vertical orientation than their horizontal dimension are said to ‘stand’ (e.g. fridges, antbeds).³ ‘Sit’ is the only posture verb in Ngan’gityemerri available for human figures in existential predication, distinguishing unmarked stance existential meanings from those entailing a postural implication (Reid

³Again, similar observations regarding copula choice (with small differences) have been made for a number of other Australian languages such as Yankuntjatjara and Pitjantjatjara (Goddard & Harkins 2002: 214–223), Gooniyandi (McGregor 1990: 310), Warlpiri (Simpson 1991: 215) and Kunbarlang (Kapitonov 2021: 306–308).

2002: 247–248). Leveraging the relation between the axial geometry of the figure object and the chosen posture verbs, the choice of posture verb can also serve to narrow the possibilities of intended senses of a polysemous nominal denoting the figure. In the Arrernte example in (3), the sense ‘stick’ is selected from possible denotations of *arne* ‘stick, tree’ because the former referent is pragmatically more associated with ‘lie’, whereas in (4) the latter sense ‘tree’ is most applicable because trees prototypically ‘stand’. The use of posture verbs to narrow denotational meanings has also been described as a posture verb function in a role that is highly reminiscent of noun classifier systems (see Aikhenvald 2000).

- (3) Arrernte (Wilkins 2006: 32)

Arne *yanhe-Ø* *arlpentye inte-me.*
tree/stick that(mid)-NOM tall/long **lie-NPP**
 ‘That stick is long.’ (lit. ‘That stick lies long.’)

- (4) Arrernte (Wilkins 2006: 32)

Arne *yanhe-Ø* *arlpentye tne-me.*
tree/stick that(mid)-NOM tall/long **stand-NPP**
 ‘That tree is tall.’ (lit. ‘That tree stands long.’)

Finally, posture verbs have also been identified as part of some specific diachronic pathways, namely in the emergence of verbal aspect—a process that has also been identified in languages outside of Australia (Kuteva 1999; Newman 2002b,a). Most of the evidence for this within the Australian context primarily comes from languages spoken in the north of the continent. Specifically, Reid (2002: 239) has identified that in serialised verb constructions, “stance verbs are, at least to some extent, bleached of posture/stance meanings, and typically convey ‘continuous’, or ‘progressive’ aspect”. Relevant languages that show evidence of this include: Ngan’gityemerri, Emmi (Western Daly, [amii1238]), Djapu (Yolngu, [djap1238]), Garrwa (Garrwan, [gara1269]; Mushin 2012: 299) and various Maningrida languages (see also a Gunwinyguan language spoken in the Maningrida region, Kunbarlang: Kapitonov 2021: 284–288). In the Diyari language (Pama-Nyungan, [dier1241]) spoken in the south of the continent, an auxiliary verb *parra-* homophonous with a verb ‘to lie’ is used for recent past tense (Austin 1981: 95).

In sum, the existing literature provides ample indication that posture verbs have a wide range of functional capabilities within and across Australian languages when occurring in locative and existential contexts. The present study seeks to provide a detailed comparison of such variation as can be found in three

languages for which posture verbs have been minimally described: Kune (Gunwinyguan), Kukatja (Pama-Nyungan) and Wik-Mungkan (Pama-Nyungan).

2 Methodology

2.1 Languages in the study

The genetic groupings of the languages in the study are shown in Table 1. Kukatja and Wik-Mungkan are both members of the widespread Pama-Nyungan language family but are spoken in the geographically disparate regions of Cape York (the far northeast of the continent) and the Tanami Desert region of Western Australia. They are only distantly related within the Pama-Nyungan family and are members of the Western Desert and Middle Paman subgroups respectively. Kune is a member of the Gunwinyguan language family, a wholly unrelated family spoken in Arnhem Land (Figure 1).

Table 1: Genetic affiliations of the focus languages

Language	Glottocode	Language family	Sub-grouping
Kukatja	kuka1246	Pama-Nyungan	Western Desert
Wik-Mungkan	wikm1247	Pama-Nyungan	Middle Paman
Kune	gune1238	Gunwinyguan	— (Variety of Bininj Kunwok)

The language sample was chosen for both novelty and convenience. All three languages, while having been described to varying levels of detail (see §2.2), have not been the focus of any systematic study with respect to copula predication or the role of posture verbs in locative or existential predication. Typologically, all three languages are still being acquired as first languages by children. Importantly, the three authors have conducted extensive linguistic documentation within the relevant speech communities and have had the opportunity to carefully elucidate posture verb semantics via dedicated elicitation techniques (§2.2), going beyond what is typically used for the basis of description in published grammars. The three survey languages all share the property of making frequent use of posture verbs but vary along several dimensions with respect to how these posture verbs feature in locative-existential clause types and other types of copula constructions. Typologically, all three languages exhibit a number of properties characteristic of non-configurational languages, especially with respect to grammatically free word order, and elision of nominal arguments. While Wik-Mungkan and Kukatja are overwhelmingly suffixing in both nominal and verbal



Figure 1: Map of languages in the study

morphology, Kune, as a non-Pama-Nyungan language, has both prefixing and suffixing morphology and a significantly more complex verbal word. Argument relations in Wik-Mungkan are predominately encoded by case-marking on dependents while Kune predominately encodes argument relations within the verbal complex. Kukatja is distinct from both in exhibiting a mixed system of both dependent marking but also relying on cross-referencing of arguments in a second position clitic complex.

Of relevance to the present study are the commonalities and differences with respect to the organisation of the class of ‘verbs’ in each language. Kukatja has two classes of lexical items which are appropriately considered ‘verbal’ in nature. One lexical type (termed *inflecting verbs* or just *verbs*) is a small, closed class of lexemes that take TAM inflection according to four conjugation classes and whose members encode relatively generic or schematic semantic content. Another lexical type (termed *preverbs*) is a larger open class of lexemes which do

not inflect for TAM values but whose members may have quite specific verbal semantic content and do not share distributional or inflectional properties of nominals or adverbs. These two classes of verbal elements frequently combine to form a type of complex predicate, sometimes termed a *preverb* or *coverb* construction (see Osgarby & Bower 2023 for a recent overview). Lexical items with postural semantics occur in both the ‘preverb’ and ‘verb’ word classes within Kukatja.⁴ By contrast, Wik-Mungkan has just one major verb class of which the posture verbs constitute members and does not make use of complex predication. For Kune, unlike the two Pama-Nyungan languages, verbal words are more polysynthetic in their structure, requiring – at minimum – affixation of pronominal arguments and TAM inflections. The verb stem occupies but one slot in a complex verbal template which further permits nominal arguments to be incorporated into the verbal structure. Additionally, Kune has several serial verb constructions which may be recruited in aspectual or spatial functions.

2.2 Data and analysis

In all three languages, we analysed data from both existing and novel material. For these languages, we consulted the existing descriptions and analyses put forward by: Peile (1971) and Valiquette (1993) for Kukatja; Evans (2003) for Kune; and Kilham et al. (1986) and Ray (2021) for Wik-Mungkan. Novel data were also collected with a specific focus on postural semantics and locative clauses, all in collaboration with language experts over the course of 2020–2023. These included both direct elicitation as well as broader corpus data (mainly monologic narratives). As part of our elicitation methods, we employed the tools of DELOCA (Guerrero 2022) and the Topological Relations Picture Series (TRPS) (Bowerman & Pederson 1992). In DELOCA elicitation, consultants are shown videos and/or photo stimuli and are asked to describe either: where a selected object is located, or what action a person is performing. In many cases, consultants were asked to judge whether the use of more than one posture verb was admissible for a given scene. In TRPS, consultants are shown line drawings on cards and asked to explain where a figure is located. The use of these standardised, picture-prompted elicitation techniques allows for crosslinguistic comparison—see, for example, the descriptions in Levinson & Wilkins (2006a), or the typology presented by Ameka & Levinson (2007). In each language, we conducted these tasks with at least four different speakers. Their responses were transcribed and coded according to the stimulus type (animacy of figure/ground, number of figure), and

⁴Inflecting verbs and/or complex verbs also occur in serialised constructions.

response (type of verb used, reduplication). Table 2 is an overview of the participants from whom data were collected using the standardized tools.⁵

Table 2: Standardised elicitation tasks probing locative-existential predication

Language	DELOCA	TRPS
Kukatja	4 speakers (2 male, 2 female)	5 speakers (3 female, 2 male)
Kune	5 speakers (3 female, 2 male)	10 speakers (8 female; 2 male)
Wik-Mungkan	4 speakers (3 male, 1 female)	7 speakers (4 male, 3 female)

We supplemented these core data with additional spatial elicitation tasks in order to examine postural use in a slightly richer array of contexts. These additional data included: responses to the Frog Story (Mayer 1969); the Man and Tree task (cf. Knudsen et al. 2023; Levinson et al. 1992; Ennever et al. 2024; Terrill & Burenhult 2008; Ennever 2024); and the Rotating Scene Machine (Blythe et al. 2022). Where a posture verb was present, these utterances were included and coded in line with the DELOCA/TRPS data.

In the following section, we introduce the basic properties of posture verbs in each language and compare their semantic range (§3.1). In §3.2, we explore how posture verbs are used in the predication of location-existence, demonstrating how posture verb choice is sensitive to ontological properties of the figure; and how individual posture verbs are used in non-locational copula constructions. Discussion of the ambiguity between locative and existential meanings is withheld until §3.3. The results from the frequency analysis of posture verb usage across the standardised elicitation tasks are presented in §3.4.

⁵The three communities that contributed language data all share internal variation amongst their populations with respect to multilingualism. Generally, all communities have exposure to Standard Australian English but vary with respect to additional linguistic influences such as the degree of multilingualism in additional Australian Aboriginal languages or their degree of utilisation of contact varieties such as Kriol [krio1252] (see Dickson 2023). Data collection in each location focussed on just the single target language. Fully accounting for community-internal variation was not the aim of data collection or of this chapter.

3 Posture verbs in Kukatja, Kune and Wik-Mungkan

3.1 Lexical semantic range

3.1.1 Lexical polysemy

The forms that constitute the basic posture verbs in each language are shown in Table 3. The Kukatja and Wik-Mungkan forms are given in their uninflected citation forms while the Kune forms are given in their verbal root forms.⁶

Table 3: Forms of the basic posture verbs in Kukatja, Kune and Wik-Mungkan

Language	‘sit’	‘stand’	‘lie’
Kukatja	<i>nyina-</i>	<i>ngara-</i>	<i>ngarri-</i>
Kune	<i>-ni</i>	<i>-di</i>	<i>-yo</i>
Wik-Mungkan	<i>nyiin-</i>	<i>than-</i>	<i>wun-</i>

Posture verbs in all three languages belong to a lexical class of (intransitive) verbs. The core semantics of the ‘sit’, ‘stand’ and ‘lie’ verbs are generally consistent across the three languages, particularly when it comes to the postural meanings they ascribe to human figures. The semantic consistency of the compared verbs mirrors a general tendency to encode a threeway ‘sit’–‘stand’–‘lie’ distinction, which Newman (2022: 1) considers to constitute “primary postures” or “basic, recurring features of human behaviour.”⁷

All the languages allow their posture verbs to be used without any accompanying locative expression simply to predicate the postural disposition of an animate subject NP. Examples from Wik-Mungkan of these types of expressions involving animate figures are presented in (5)–(7).

⁶Across the three languages, only the *nyiin-* and *nyina-* forms in Wik-Mungkan and Kukatja respectively have been demonstrated to be cognate, descending from a common proto-Pama-Nyungan form **nyi:na* (Alpher 2004: 509–510).

⁷The three languages do differ with respect to the form and/or derivation of dynamic posture predicates, used to describe the act of assuming a particular posture, i.e. predications of the kind realised in English by the expressions ‘stand up’, ‘sit down’, and ‘lie down’ but this is beyond the scope of this study. For language particular descriptions of this aspect of posture verbs, see Ennever (2024: 181–184) for Kukatja, Ray (2021: 310–312) for Wik-Mungkan, and Evans (2003: 379–380) for Kune.

- (5) Wik-Mungkan⁸ (Michella Pahimbaung; deloca_mp_230610)
Than~than-pul-a.
 RDP~stand-3DU.PRS-a
 ‘The two are standing.’
- (6) Wik-Mungkan (Maree Kalkeeyorta; navigation-task_mk_230716)
Waath-a nyiin-Ø.
 crow-a sit-3SG.PST
 ‘A crow sat.’
- (7) Wik-Mungkan (Rufus Namponan; deloca_rn-rw_230610)
Nil wun~wun-Ø, a’ ek’-Ø ngul.
 3SG RDP~lie-3SG.PST and get.up-3.SG.PST then
 ‘He was laying down, and then he got up.’

Even though the verbs for ‘sit’, ‘stand’, and ‘lie’ can be used in a corresponding fashion when predicating posture in all three languages, they pattern differently in terms of their semantic extensions. The distribution of polysemies across the class of verbs in the three languages is illustrated in the crosslinguistic semantic map shown in Figure 2. Note that the semantic map includes an aspectual function (DURATIVE) specific to the ‘sit’ verb in Kune. This role is restricted to its occurrence in a type of aspectual serial verb construction of the kind mentioned in §1.2—a function not discussed further in the present paper (see Evans 2003: 659 for Kune-specific details).

The semantic extensions identified here are not unusual for Australian languages and broadly align with many of the experientially-motivated semantic extensions identified by Newman (2022: 10–11). The ‘sit’ verbs tend to exhibit extensions invoking a durative component of the kind ‘dwell’, ‘reside’ which Newman hypothesises as a reflection of ‘sitting’ as a kind of “default” posture (Newman 2022: 10). Kukatja and Kune, for example, follow this cross-linguistic tendency, utilising their respective ‘sit’ verbs to mean ‘live/reside’ (8 and 9) or ‘stay/remain’ (10). By contrast ‘stand’ verbs are typically excluded from such extensions and are instead used for expressing temporary location.

- (8) Kukatja: reside (TKU: PEILE_A06-001932A: 1393.1)
Tjitja turru pirti-ngka nyi-nin.
 IDENT bird hollow-LOC sit-PRS
 ‘This bird [the *luurn*] lives in hollows.’

⁸Word final *-a* is generally analysed as ‘rhythmic juncture’ in Wik-Mungkan (see Ray 2021: 56–57 and references therein) although its precise function is not well understood, nor agreed upon by analysts. Here we gloss it rather agnostically as ‘-a’.

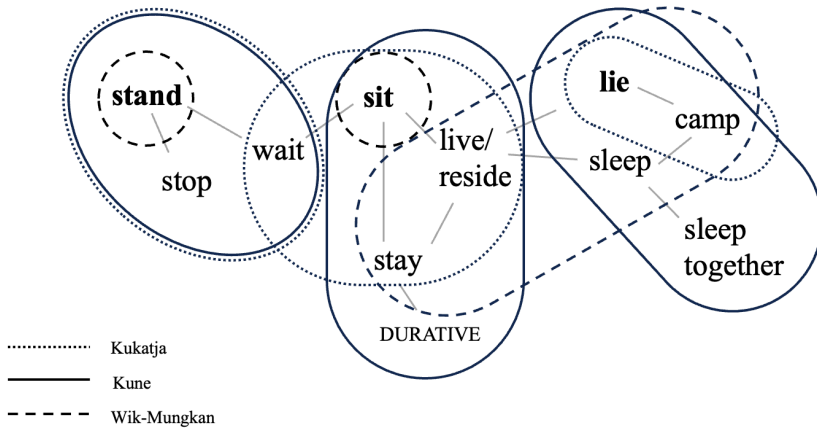


Figure 2: Comparative semantic map of ‘sit’, ‘stand’ and ‘lie’ verbs in Kukatja, Kune, and Wik-Mungkan

- (9) Kune: reside (Kamanj Michelle Martin; KLLC_20211004a)
Kondanj ngarri-ni bad kondanj nga-djordm-inj, Kune
 here 1AUG.EXCL-sit.NPST CONJ here 3MIN-grow_up-PST Kune
darrme.
 LOC
 ‘We live here but I (also) grew up here, on Kune country.’
- (10) Kukatja: stay/remain (Peile 1971: 20)
Ya-nku-marranpa=rna, langa-kutjupa-rri-ngu=rna
 go-IRR-AVERT=1SG.SBJ ear-other-INCH-PST=1SG.SBJ
nyi-nama=lpi=rna.
sit-IPFV=then=1SG.SBJ
 ‘I was going to come, but I changed my mind (lit. ‘I became the other ear’)
 and so I stayed.’

In Wik-Mungkan, however, the sense ‘live/reside’ is a feature of the ‘lie’ verb, *wun* (11). Like ‘sitting’, ‘lying’ postures are similarly able to be maintained for a long period of time and are experientially associated with the body at rest.

- (11) Wik-Mungkan: reside (Kilham et al. 1986: 240)

Wachan wun-an thannt=am.

long_way lie-3SG.PRS 3PL.DAT=ABL

‘(He) lives a long way from them.’

In both Wik-Mungkan (12) and Kune (13), the ‘lie’ verb is polysemous with a sense of ‘sleep’, again reflecting intuitive experiential associations.

- (12) Wik-Mungkan: sleep (Perry Yunkaporta; space-elicitation_py_230411)

My cousin nan wun-Ø.

my cousin MED lie-3SG.PST

‘My cousin slept there.’

- (13) Kune: sleep (Evans 2003: 156)

Karri-yo ku-kawadj.

1AUG.INCL-lie.NPST LOC-sand

‘We sleep on the sand.’

Interestingly in Kukatja, there is no equivalent extension of *ngarri-* ‘lie’. Instead, a separate verbalised predicate *nyarli-rri-* ‘sleep-INCH’ (from a nominal predicate *nyarli* ‘(a)sleep’) is used to refer to the activity of sleeping (14). Nevertheless, *ngarri-* ‘lie’ does extend to the narrower sense of ‘(a night’s) camp’, or ‘camp overnight’, a usage that features heavily in travelogues, such an example is shown in (15). In Kune and Wik-Mungkan the semantic space of ‘lie’, ‘sleep’ and ‘stay/camp overnight’ is covered by just the one posture verb, ‘lie’.

- (14) Kukatja: ‘sleep’ (ALL: TEN2-2022_025-03: 1069.800)

Ngatja kunyarr-pa nyarli-rri-n waru-wana.

PROX dog-EP sleep-INCH-PRS fire-PERL

‘This dog is sleeping by the fire.’

- (15) Kukatja: ‘camp (overnight)’ (MSK: Manungka_01-035158: 1839.553)

Tjii-tjanu:: Laka-kutu. Laka-ngka=latju ngarri-rnu.

IDENT-ABL Laka-ALL Laka-LOC=1PL.EXCL.SBJ lie-PST

‘After that (we went) to Laka. We stayed/camped overnight at Laka.’

Lastly, both ‘sit’ and ‘stand’ verbs in Kukatja can be used with the sense ‘wait (for)’ in combination with a dative argument as in (16) and (17), and the ‘stand’ verb has an additional lexical sense: ‘stop’. In Kune, the ‘stand’ verb can similarly mean ‘stop’ or ‘wait’ (see, e.g., Evans 2003: 685). Neither of these senses are covered by any posture verb in Wik-Mungkan and Wik-Mungkan is further notable for not demonstrating any clear instances of polysemy for the ‘sit’ member.

- (16) Kukatja: ‘wait’ (TKU: PEILE_A06-001932A: 1393.1)
Nyi-ninpa=rna=ra tjarlu-ku.
 sit-PRS=1SG.SBJ=3SG.D big-DAT
 ‘I am waiting for the boss.’
- (17) Kukatja: ‘wait’ (NKI: PEILE_A06_001933B: 581.774)
Wiya=tju=n ngara-ku!
 NEG=1SG.D=2SG.SBJ stand-POT
 ‘Don’t wait for me!’

3.1.2 Posture verbs and wider systems of postural specification

Before proceeding with the analysis of posture verbs in locative-existential and other copula constructions, we first provide a brief illustration of how the three survey languages express postures other than ‘sit’, ‘stand’ or ‘lie’. Here we find quite a diversity of strategies across the verbal and nominal domain. However, barring a single exception (*werndi* ‘be elevated’ in Kune), none of these strategies described are standardly recruited in locative-existential functions (without a specifically postural meaning), nor in copula functions. Within the verbal domain, Kune is notable for having several morphologically complex verb stems that involve the posture verb thematics, especially *di* ‘stand’, as in *djab-di* [upright-stand] ‘be upright’ (18) or *barddurrungkulh-di* [ground_with_knees-stand] ‘kneel’ (19)–see also Evans (2003: 339).

- (18) Kune (Belinj Jay Jurrupula Rostron: KLLC_20220807h)
Kun-dulk ka-djab-di.
 IV-tree 3MIN-upright-stand.NPST
 ‘The tree is upright.’
- (19) Kune (Bulanj Russell Brian: KLLC_20230621a_RuB)
Kabini-barddurrungkulh-di.
 3UA-ground_with_knees-stand.NPST
 ‘They are kneeling.’

In (18), the postural meaning of ‘be upright’ is encoded by the combination of verbal preponent *djab-* and verbal thematic *-di* while in (19), the postural meaning of ‘kneel’ is compositionally formed via the stand verb *-di* in combination with the lexicalised verbal preponent *barddurrungkulh-*, involving the morpheme *bard-* ‘knee’. The Kune complex verb *-werndi* ‘be elevated’ (18) will be returned to in §3.4 since it alone is recruited in copula function and is even treated as a ‘stance’

verb by Evans (2003: 339), despite its semantics being primarily positional rather than postural.

Kukatja has both complex verbs, involving postural preverbs (e.g. *wartu-* ‘prostrate’), as well as secondary predicate and adjunct based strategies. Examples are shown in (20)–(22).

- (20) Kukatja (PSM: TEN2-2023_003-01: 573.796)
(Tutju lamparn manga) wartu-ngarri-npa.
 woman small girl prostrate-lie-PRS
 ‘The little girl is lying face-down.’
- (21) Kukatja (MMN: TEN2-2023_004-01: 676.227)
Ngatja tutju nyina-ti-n karlitjalpi.
 PROX woman sit-ACT-PRS cross_legged
 ‘This woman is sitting down cross-legged.’ (*karli.tjalpi* lit. ‘boomerang awkward’)
- (22) Kukatja (GLE: TEN2-2023-001-02: 98.1)
Tutju lamparn-pa murti-kurlu ngari-n.
 woman small-EP knee-PROP stand-PRS
 ‘This young woman is kneeling (lit. ‘standing with knees.’).’

In (20), the complex postural meaning of ‘lying face-down’ is encoded in the preverb-verb combination *wartu-ngarri-*. In (21), a nominal *karlitjalpi* ‘cross-legged’ is used as a resultative secondary predicate assigning a resultant postural state to the subject figure brought about by the dynamic main predicate *nyina-ti-* ‘sit down/become seated’. Finally in (22), a proprietive case marked nominal expression, *murti-kurlu* ‘knee-PROP’, does not invoke a posture as a resultant state of a dynamic predicate but rather implicates a body part (specifically ‘knees’) as being involved in the postural event of the main verb *ngari-* ‘stand’, an idiomatic way of expressing ‘kneeling’.⁹

Wik-Mungkan is a little more restricted than either Kune or Kukatja in its lexical and constructional resources for specifying particular postures but exhibits a similar strategy to the body part strategy of Kukatja shown in (22), through case-marked body parts as shown in (23). Specifically, the body-part *pungk* ‘knee’ is case-marked with the polyfunctional instrumental/locative case suffix and has the syntactic role of an adjunct modifier, external to the verb phrase.

⁹Propriptive (20) and instrumental (21) case-marked NPs in many Australian languages have (secondary) predicative functions as depictives wherein the case-marked nominal serves to characterise a property of the subject NP throughout the duration of the verbal event.

- (23) Wik-Mungkan (Rosalie Wolmby; deloca_rn-rw_230610)
Nil pungk=ang nyiin-an.
 3SG.NOM **knee=INS/LOC** sit-3SG.PRS
 ‘He is kneeling/sits with knees.’

3.2 Posture verbs in the predication of location and existence

3.2.1 Overview of posture verbs and construction

In all three languages, the posture verbs ‘sit’, ‘stand’ and ‘lie’ are used as the predicative element in locative-existential constructions, coding a spatial relation between a figure and a ground, i.e. ‘be (at)’. While all three languages pattern similarly with respect to the use of posture verbs in this function with animate figures (§3.2.2), the languages show differences when one considers inanimate figures (§3.2.3) and their use in copula constructions beyond locative-existentials (§3.2.5). This information is shown in Table 4. Note that in the final column ‘copula functions’ refers to contexts in which no locative-existential meaning is present and the copula links a subject with a (non-spatial) complement.

The key differences in Table 4 relate to the choice of posture verb that is recruited both as a default verb (in locative-existentials) and in fully grammaticalised copula function in other copula constructions. For Kukatja the member is *nyina*- ‘sit’, for Kune it is *-di* ‘stand’ and for Wik-Mungkan, it is *wun*- ‘lie’.

3.2.2 Locative-existential predications with animate figures

For all three languages, when the figure referent is animate, each of the core posture verbs can be used to describe a postural configuration that in some ways aligns with the human prototype. Contrastive examples of posture verbs with animates were given for Wik-Mungkan in (5)–(7) and we provide additional examples of such stance-focussed predicates for the three languages in (24)–(26). In each example, the focus of the proposition is that the figure subject is in a specific posture.

- (24) Kukatja (Ennever 2024: 164)
Ngayunpa=la=nta nyi-nin kinti nyuntu-ngka.
 1PL=1PL.EXCL.SBJ=2SG.OBJ/L **sit-PRS** close 2SG-LOC
 ‘We are sitting close to you.’

Table 4: Functions of basic posture verbs across both construction and figure type

Language	Posture verb (gloss and function as stance predicate)	Locative-existential constructions		Copula functions
		Animate figure	Inanimate figure	
Kukatja	<i>nyina</i> ‘sit’	‘be in seated position (at)’	‘be (at)’ (unmarked for stance)	COP
	<i>ngara</i> ‘stand’	‘be in vertical position (at)’	‘be (at)’ (of vertical figure)	–
	<i>ngarri</i> ‘lie’	‘be in horizontal position (at)’	‘be (at)’ (of horizontal figure)	–
Kune	<i>-ni</i> ‘sit’	‘be in seated position (at)’	<i>RARE</i>	–
	<i>-di</i> ‘stand’	‘be in upright or vertical position (at)’ (supported by legs) ‘be (at)’ (unmarked for stance)	‘be (at)’ (unmarked for stance)	COP
	<i>-yo</i> ‘lie’ ^a	‘be in horizontal position (at)’	‘be (at)’ (of horizontal figure)	–
Wik-Mungkan	<i>nyiin</i> ‘sit’	‘be in seated position (at)’ (also crouching with support from feet)	<i>RARE</i>	–
	<i>than</i> ‘stand’	‘be in upright position (at)’	‘be (at)’ (of upright figure)	–
	<i>wun</i> ‘lie’	‘be in horizontal position (at)’ ‘be (at)’ (unmarked for stance)	‘be (at)’ (unmarked for stance)	COP

^aA fourth Kune verb *wərndi* ‘be up high’ (*barndi* in related dialects) is included by Evans (2003: 339) as a posture verb. It is not included in the table here since our focus is primarily on the semantically defined set of three (sit, stand and lie). However, see §3.4 for its quantified use in the experimental tasks.

- (25) Kune (Bulanj Simon Brian; KLLC_20220709b)
Kun-rak-kah ka-ni~ni.
 IV-fire-LOC 3MIN-RDP~sit.NPST
 ‘He (the man) is sitting by the fire.’
- (26) Wik-Mungkan (Kay Tybingoompa; trps_kt_230416)
Ku’waak pek~pek=ang=an table=ang=an nyiin-Ø.
 cat RDP~below=LOC=DEF table=LOC=DEF sit-3SG.PST
 ‘A cat sat under the table.’

In examples (24)–(26), the respective ‘sit’ verb in each language is used in an expression of a locational relation. In all the languages, the chosen verb is reflecting an actual postural configuration of the figure. We can contrast, for example, the sitting posture of the animate figure ‘the cat’ in (26) with the description of a dog lying down inside a dog house in (27).

- (27) Wik-Mungkan (Perry Yunkaporta; trps_py_221116)
Ku’=an awuuch many=ang=an wun-Ø-a.
 dog=DEF house small=LOC=DEF lie-3SG.PST-a
 ‘The dog lay inside the small house.’

The distinct postural specification of the dog in (27) as compared to that of the cat (26) is accomplished through the use of a contrastive postural verb *wun-* ‘lie’ rather than *nyiin-* ‘sit’. In fact, in all three languages, this is the preferred method for contrasting basic ‘sitting’, ‘standing’ and ‘lying’ postures for all animate figures (not just humans).

3.2.3 Locative-existential predications with inanimate figures

Moving to inanimate figures, the situation is a little more complex. All three posture verbs are attested in use with inanimate figures in each language. However, each language differs with respect to which posture verb is most commonly used: the preferred member is *nyina-* ‘sit’ in Kukatja, *-di* ‘stand’ in Kune, and *wun-* ‘lie’ in Wik-Mungkan—see examples (28)–(30). In what we might call a ‘default’ context, the posture verb primarily serves as a predication element in the locative construction. In Kukatja and Wik-Mungkan, the verb simply provides a host for TAM marking as shown in (28) and (30). In Kune, with its more polysynthetic word structure, the posture verb can oftentimes combine with an incorporated nominal expressing the ground relation—such as *kolh(-no)* ‘water’ in (29)—and further hosts a prefixed subject marker expressing the figure.

- (28) Kukatja (ALL: TEN2-2022_022-01: 913.08)
Sand-pa=lu nyi-nin motika-ngka.
 sand-EP=SG.L sit-PRS car-LOC
 ‘Sand is in the vehicle.’
- (29) Kune (Bulanj Simon Brian; KLLC_20220711b)
Ka-kolh-di.
 3MIN-water-stand.NPST
 ‘The water is in there.’
- (30) Wik-Mungkan (Beulah Pootchemunka; trps_bp_230412)
Cup=an an=an table=ang=an wun-Ø.
 cup=DEF DIST=DEF table=LOC=DEF lie-3SG.PST
 ‘The cup was on that table.’

The question that arises from data such as (28)–(30) is to what extent the ‘posture’ verbs being used here represent semantically-contentful predicates or whether they can be said to be entirely bleached of their semantic makeup. We find evidence in all three languages that there is not a simple categorical answer to this but rather each of the verbs can be used with varying levels of semantic-specificity along a cline from ‘stance-specified intransitive verbs’ through to ‘semantically null’ or ‘dummy’ copula. We also find evidence that these items are likely interpreted at the pragmatic level based on properties of the figure, properties of the ground, use of gesture, and the wider discourse context.

We first justify this position by demonstrating that the default posture verbs can be used at ends of this functional continuum. On one end, we have already seen that with animate figures, the posture verbs are used contrastively to assert human or human-like postures (see Wik-Mungkan examples in (5)–(7) and §3.2.2). At the other end of the spectrum, all three languages at least show some evidence of posture verbs being used in contexts where their postural semantic content is at odds with the postural properties of the figure being located (posture verbs used in non-locative copula constructions are also described in §3.2.5). This constitutes reasonable evidence that such terms can be used, not in their fully semantically specified sense in locative-existential predication, but in a purely relational role. In (31), a Wik-Mungkan speaker responded to the pictorial prompt of an upright bottle on a table with a locative expression involving the “default” posture verb *wun-* ‘lie’. We can say here that the role of *wun-* is simply to serve as the predication element ‘be (at)’ in a locative construction.

- (31) Wik-Mungkan (Perry Yunkaporta; space-elicitation_py_230411)
Chang=an-a wonk-pal kuuw an-a wun-Ø.
 water=DEF-a side-CIS.LOC west DIST-a lie-3SG.PST
 ‘The water bottle is towards that west side.’

Default uses of copula verbs such as *wun-* in (31) are not uncommon with inanimate figures in Wik-Mungkan. While such default uses rarely occur with animate figures, they do crop up in elicitation and wider corpora. An example of the stance meaning being bleached in the context of predicating location of an animate figure in Wik-Mungkan is shown in (32), describing the scene in Figure 3.



Figure 3: Picture 34 from the Topological Relations Picture Series (Bowerman & Pederson 1992)

- (32) Wik-Mungkan (Beulah Pootchemunka; trps_bp_230412)
Keny=ang wun-an-a ngeen=ang=an, than~than-an.
 above=LOC lie-3SG.PRS-a something=LOC=DEF, RDP~stand-3SG.PRS
 ‘(The man) is (lit. lies) on top of the something, (he) is standing.’

In the initial main clause in (32) *wun* ‘lie’ is used in copula function, while in the prosodically separated clause, (reduplicated) *than* is used to assert the actual posture (‘stand’) of the man being located on top of the building. This kind of flexibility in the deployment of posture verbs across the postural-locational semantic space was also revealed in meta-discussion with speakers in the course of elicitation. For example, following the elicitation of the copula use of *wun-* in (31), later conferral with the speaker revealed that *than-* ‘stand’ was still considered equally appropriate for such a locative scene, but instead involved a shift in focus to the bodily (or more accurately ‘geometric’) stance of the figure itself (in line with characterisations represented in Table 4). This shows how the bleaching of the postural semantics of *wun-* ‘lie’ verb does not preclude a speaker from using

a non-prototypical posture verb in order to emphasise, contrast or pick out a certain property of the figure. In other words: while movable inanimate objects are generally described with *wun-* ‘lie’ in locative expressions in Wik-Mungkan, if *than* ‘stand’ is used, the pragmatic consequence is a focus on an upright configuration. Very similar situations were found to hold for both Kukatja and Kune. In Kune, while *-di* is the unmarked posture verb in locative-existential predication involving an inanimate figure, we nevertheless find data of the kind shown in (33) where *-yo* ‘lie’ is used instead.

- (33) Kune (Bulanjdjan Risselle Brian; KLLC_20210917f)
Spun ka-yo~yo kanjdji kun-madj.
 spoon 3MIN-RDP~lie.NPST under IV-cloth
 ‘The spoon is (lying) underneath the napkin.’

Remarking on data of the kind shown in (33), Evans has made the following parallel observations regarding locative-existential clauses in Kune, emphasising the flexibility to use other (non-default) copulae in the language, such as *barndi* ‘be high’:

There is some room to stress salient facts of the situation: normally one says of fruit in a tree that are not too high up, *kamimdi* [it-fruit-stands] ‘there are fruit’, but if one wanted to stress how high up they were one could say *kamimbarndi* [it-fruit-be_high] ‘there are fruit up high’. (Evans 2003: 561)

Such flexibility in the deployment of posture verbs in locative-existential constructions has been noted cross-linguistically. As Newman (2022: 7) describes, “the use of the posture morphemes in locative/existential constructions may not completely displace the original posture meaning”. This is also the case in the languages investigated here. A working hypothesis is that the chosen posture verb unmarked for stance in locative predication types is indicative of a general level of bleaching of the given verb. This would result in the demands of the figure object semantics being less strict. Indeed, as we shall see, the bleached verb is the one used for copula predication in which a postural meaning is entirely absent for two of the languages.

3.2.4 Posture verb choice and polysemous expressions for the figure

While we saw in the previous sections that posture verb selection with inanimates show some flexibility, there is one specific context in which we find some

evidence that posture verb choice has a contrastive, semantic role. This is primarily in the disambiguation of polysemous nominal expressions (as noted in the Australian context by Wilkins 2006 for Arrernte, for instance). This function has only been observed for scenes involving inanimate figure subjects. For example, the polysemous noun *yuk* in Wik-Mungkan can both mean ‘tree’, ‘stick’, and is also a generic classifier noun for ‘stuff’. Examples (34) and (35) demonstrate how the posture verbs can be used to align and thus disambiguate the sense-type of *yuk* that is being described.

- (34) Wik-Mungkan (Dennis Wolmby; deloca_dw_230620)
Yuk many an=an bowl al=ant=ang=an **wun~wun-Ø**.
 tree small DIST=DEF bowl DIST=DAT=LOC=DEF **RDP~lie-3SG.PST**
 ‘That little stick was lying in the bowl there.’
- (35) Wik-Mungkan (Kay Tybingoompa: trps_kt_230416)
Yuk=an an=an-a **than-Ø** yoik=ang=an-a.
 tree=DEF DIST=DEF-a **stand-3SG.PST** mountain=LOC=DEF-a
 ‘That tree stood on the mountain.’

Here, *than-* ‘stand’ is used for a vertical inanimate figure, while *wun-* ‘lie’ is used for a horizontal figure. *Wun-* ‘lie’ is the posture verb appropriate for any inanimate figure and especially one arranged or distributed horizontally on the ground. These examples demonstrate one function of the posture verb selection. The ‘classifying function’ of posture verbs evident in contrastive examples like (34) and (35) relies on the fact that the posture verbs themselves are not completely bleached of their semantic content in locative-existential function but can be sensitive to geometric properties of inanimate figures.

3.2.5 Beyond locative-existential predication

All three languages have just one posture verb which is used as a copula in construction types that do not encode either existence or location. These represent the most ‘bleached’ functions of these lexical items since the postural semantics are wholly absent and the copula simply serves to link a subject relation to a subject complement. The argument structure of the copula construction in these cases involves two unmarked (absolute) arguments, or it may involve an unmarked subject and another argument in a semantic case. Examples are (36)–(37).

- (36) Kune (Bulanj Russell Brian; KLLC_20230621a_RuB)

Laid on ka-di.

light on 3MIN-be.NPST

‘The light is on.’

- (37) Kukatja (Greene et al. 1992: 27)

Ngaanpa=ya lukuti-yuru nyi-nama.

3PL=3PL.SBJ grub-SEMBL be-IPFV.PST

‘These (men) were like witchetty grubs.’

In (36), the underlined elements represent codeswitches from Kriol within a Kune matrix clause. The nominal stative predicate *on* ‘on’ combines with the main clause predicate, the Kune verb *-di* ‘be (stand)’, expressing the stative predicate ‘be on’. The subject of the predication is the Kriol-sourced *laid* ‘light’ which shows agreement with the Kune verb via the third person minimal subject prefix *ka-*. The clause lacks any invocation of postural semantics and is best analysed as involving a ‘copula’ use of *-di*. In the Kukatja example in (37), the predication head is the semblative case-marked nominal expression *lukuti-yuru* ‘like witchetty grubs’ which is linked via the copula *nyi-nama* ‘be(sit)-IPFV.PST’ to the subject relation instantiated by the demonstrative pronoun *ngaanpa* ‘these ones’. The subject is further cross-referenced by the second position subject enclitic *=ya*.

Example (38) shows a copula use of *wun-* ‘lie’ in Wik-Mungkan. The copula hosts the portmanteau of optative verbal inflection and subject person/number, specifying the subject and temporal/modal conditions of the ascriptive nominal predicate *meech* ‘(be) hungry’.

- (38) Wik-Mungkan (Kay Tybingoompa, fieldnotes_240129)

Meech wun-iyin!

hungry be-1PL.EXCL.OPT

‘We might go hungry!’

3.3 On the ambiguity of locative and existential predication

So far we have focussed on the role posture verbs play in predicating a particular stance to a figure or predicating a location to a figure. An as yet unaddressed and complicating factor is that the same types of constructions used for location are also used in the study languages for predicating ‘existence’. Here we are referring to the meaning differences realised in English between the clauses (a) and (b) in (39).

- (39) a. Location predication
The cat is on the table.
b. Existential predication
There is a cat on the table.

The terminology in the literature for distinguishing predications of the sort shown in (39) is rather fraught. Since languages vary widely with respect to their encoding of the semantic differences between (39a) and (39b), some of the typological literature has instead attempted to ground the differences not in a single structural property but instead through a distinction in cognitive perspectivisation: the former involving perspectivisation starting from the figure; and the latter involving perspectivisation starting from the ground (see Creissels 2019). Ultimately, the differences between these two relate to information packaging and the respective salience of figure and ground relations. Haspelmath (2025) argues that constructions of the kind (a) and (b) in (39) can be typologically defined on the basis of formal definite marking of the figure (in the former and not the latter) presumably since this provides a sufficient cross-linguistic metric for information structure. Haspelmath terms the constructions so defined ‘predlocative’ and ‘existential’ respectively.¹⁰ Word order (as in English) may also play a role. However not all languages encode information structure in the same way and so morphosyntactic based distinctions may (understandably) not result in meaningful distinctions in every language.

For the languages in the present survey, two of them (Kukatja and Kune) lack any dedicated marking of definiteness, although demonstratives in these languages can have determining roles amongst their various other functions (more generally specifying identifiability based on distance relations or anaphoricity).¹¹ None of the three surveyed languages show a consistent correspondence between word order and locative vs. existential interpretations. Since the posture verbs involved in these constructions can also have stance-predication functions, we are in fact left with a threeway ambiguity in the predication of stance, location and existence in two of the surveyed languages, Kune and Kukatja. The argument structures of the three closely aligned clause types are presented in

¹⁰Other analysts prefer the terms PLAIN-LOCATIONAL and INVERSE-LOCATIONAL (after Creissels 2013, 2019), reflecting the salience of the figure and ground respectively and reserve the term ‘existential’ clauses for those which do not predicate a location to a figure, i.e. wholly non-locative existential clauses of the kind instantiated by the English ‘Aliens exist’ (Haspelmath 2025 refers to these as ‘hyparctics’).

¹¹See Louagie (2016) for a typological discussion on the status of determining elements in Australian Aboriginal languages more generally.

Table 5: Argument structure of postural, locative, and existential clauses

Clause type	Argument structure		
	<Figure>	<Ground>	
stance-focussed	subject	(adjunct)	verb
locative	subject	complement ^a	(verb)
existential	subject	(adjunct)	(verb)

^aUnmarked, or marked with spatial case (e.g. locative, perlative, etc.)

Table 5. Note that for existential clauses either the verb or the locational adjunct can go unexpressed but not both. Wik-Mungkan only differs from Kukatja and Kune insofar as existential clauses can be distinguished based on the presence of definite marking of the figure in locative and stance-focussed clause-types, and lack of definite marking in existential clauses.

Since each of the three clause types only differs with respect to the obligatory realisation of specific elements, we describe the types of ambiguity in cases i) with all three arguments; ii) without a verb; and iii) without a ground relation. For Kukatja and Kune, when a posture verb, subject and spatial complement/adjunct are all expressed, there is threeway ambiguity between locative, existential and postural meanings for animates and generally two-way ambiguity for inanimates. A Kukatja example is shown in (40).

(40) Kukatja (ALL: TEN2-2022_022-01: 913.08)

- a. *Kunyarr-pa kalyu-ngka nyi-nin.*
dog-EP water-LOC sit-PRS
‘A/the dog is in the water.’ / ‘There is a/the dog in the water.’ / ‘A/the dog is sitting in the water.’
- b. *Sand-pa=lu nyi-nin motika-ngka.*
sand-EP=3SG.L sit-PRS vehicle-LOC
‘Sand is in the vehicle.’ / ‘There is sand in the vehicle.’

When a posture verb is absent, there is ambiguity between locative and existential readings. A Kune example is shown in (41).

- (41) Kune (Bulanjdjan Risselle Brian; KLLC_20220708f)
Kun-dulk konda-beh koyek, kureh-beh bininj walem.
iv-tree here-side east there-side man south
‘The tree is on this side, east; the man is on that side, south.’ / ‘There is a tree on this side, east; there is a man on that side, south.’

When a verb is present but the locative argument is absent, a ‘locative’ interpretation is of course not possible. However, in these contexts, there is still structural ambiguity between postural and existential readings as illustrated in the Kune example in (42).¹²

- (42) Kune (Bulanj Russell Brian; KLLC_20220711b)
Kunj na-wern ka-ni.
kangaroo I-many 3MIN-sit.NPST
‘There are lots of kangaroos.’ / ‘Lots of kangaroos are sitting’.

Unlike Kune and Kukatja, Wik-Mungkan allows for definite marking of nominal expressions, providing formal grounds to distinguish between locative and existential clauses, in line with widespread typological observations. An example of an existential clause in Wik-Mungkan is shown in (43).

- (43) Wik-Mungkan (mt_mk-cak_1_230613)
A’ keny=angk=an-a pam-a than-an.
and top=LOC=DEF-a man-a stand-3SG.PRS
‘And on the top (there) is a man.’

In this example, the figure relation *pam* ‘man’ is not definite marked, and is located on the top side. The ground object is the locational noun *keny* ‘top’. The ground object is definite marked and marked as a spatial adjunct by the locative case. This example is an extract from a Man and Tree game, and was preceded by a description of the tree’s location. In this context then, the new information is the existence (and location) of the man. However, beyond definite marking, the same flexibility with respect to the realisation of arguments is observed in Wik-Mungkan as found for the other two languages (44)–(47). These range, respectively, from the most minimal realisation to one in which all possible constituents are overtly realised. In (44), the minimal locative marked complement

¹²Evans (2003: 478) observes that one of the contexts in which nominals are incorporated with intransitive subjects in Bininj Kunwok languages is when they introduce new referents or assert existence. This is not the only function of such clause types, although it is a tendency observed in the data collected for Kune by the first author.

NP is expressed, describing the location of an unexpressed but discourse topical subject figure as being at the house. In (45), both the locative marked complement nominal expression (this mountain) and the figure subject NP (the tree) are expressed. In (46), the obligatory complement NP (the plate) and a posture verb functioning as a copula-like verb is expressed. Lastly, in example (47) the maximal argument structure is illustrated.

- (44) Wik-Mungkan (kt: trps_kt_230416)
Complement (locative predication)
Awuuch=ang=an-a.
house=LOC=DEF-a
‘[It is] at the house.’
- (45) Wik-Mungkan (kt: trps_kt_230416)
Subject + complement (locative predication)
Yuk=an-a yoik=ang in=an=iy-a.
tree=DEF-a mountain=LOC PROX=DEF=TOP-a
‘The tree [is] on this mountain.’
- (46) Wik-Mungkan (mp: deloca_mp_230610)
Complement + copula verb (locative/stance predication)
Plate=ang=an wun-Ø-a.
plate=LOC=DEF lie-3SG.PST-a
‘[It] was/lay on the plate.’
- (47) Wik-Mungkan (kt: trps_kt_230416)
Subject + complement + copula (locative/stance predication)
Lat in=an=iy-a table=ang than-an.
book PROX=DEF=TOP-a table=LOC stand-3SG.PRS
‘This book is/stands on the table.’

3.4 Frequency comparison of posture verbs

The existing descriptions of the use of posture verbs in locative and existential functions in at least Kune and Wik-Mungkan provide a characterisation of predictability and categoricity (see Ray 2021: 318 for Wik-Mungkan; and Evans 2003: 560–562 for Kune). Contrary to these descriptions, we find that in present day Wik-Mungkan and Kune, the use (and omission) of posture verbs are used with greater flexibility and with sensitivity to the various factors laid out in §3.2-3.3.

In this section we compare flexibility of posture verbs using results from standardised elicitation tasks introduced in §2.2.

Despite the structural possibility of posture verbs in locative-existential predications, they are not syntactically obligatory, nor are they even necessarily the quantitatively most frequent construction type for some of these predicate types in these languages. An under-researched question, then, is to what extent posture verbs are a preferred strategy amongst possible alternative construction types for encoding locative-existential meanings. In Wik-Mungkan and Kukatja (in contrast to Kune), the use of posture verbs in locative-existential contexts is a marked way of expressing these types of predicates. Instead, truly verbless locative and existential expressions are more common, as in examples (48) and (49)—a construction type for which Australian languages are reasonably well-known (Lynch 1998: 198; Simpson 2023: 227–228, Dixon 2002b).

- (48) Kukatja (GLE: TEN2-2023_001-02: 374)

Lingka lamparnpa kalyu-ngka.

snake small water-LOC

‘The small snake is in the water.’

- (49) Wik-Mungkan (Michella Pahimbaung; deloca_mp_230610)

Yoik keny=ang-a wuuch=an-a.

mountain above=LOC-a house=DEF-a

‘The house is on top of a mountain.’

In the Wik-Mungkan example in (49), the figure *wuuch* ‘house’ is located on top of the ground object *yoik* ‘mountain’ without use of a verb in a postural or copula function. Instead, the locative case-marked expression *yoik keny=ang-a* ‘on top=LOC’ itself serves as the sole predication element within the clause. Similarly in (48), the locative case suffix *-ngka* attaches to the nominal *kalyu* ‘water’ and forms a predication head *kalyu-ngka* that does not need a linking copula to predicate a location of the figure (the subject *lingka lamparn* ‘small snake’). The ability for a locative case suffix to function predicatively is a feature observed in Australian Aboriginal languages more broadly (Simpson 2023: 227–228).

We utilised results from the two main elicitation task (DELOCA and TPRS) to provide some quantified gauge of preferences among speakers for types of locative descriptions. Results were coded as to whether they invoked a posture verb (and which member), a fully verbless strategy (as in (48) and (49)), or a miscellaneous ‘other’ category (to be exemplified shortly). A key hypothesis is that the

animacy of the figure will influence the preference for posture verb usage: specifically, referents higher on the animacy hierarchy are expected to show more frequent use of posture verbs in locative-existential predications; while referents lower on the animacy hierarchy will show more frequent usage of non-posture verb strategies. All three posture verbs are anticipated to be used frequently with animate figures while inanimates will show either: i) a general dispreference to be predicated of a location via a posture verb or, alternatively ii) a preference to use but a single (bleached) posture verb in copula function.

We first present the results for Kukatja as shown in Table 6. The Kukatja data show a reasonably robust relationship between figure type (based on a basic animacy hierarchy) and predication strategy. Posturals were relatively rarely used for inanimate figures in Kukatja, accounting for just 23% of locative descriptions while they were used for some 66% of scenes that involved an animate figure.

Table 6: Kukatja strategies for predicating location in DELOCA and TPRS

Figure type	Posture verb			Non posture verb	
	<i>nyi-</i> 'sit'	<i>ngara-</i> 'stand'	<i>ngarri-</i> 'lie'	Verbless	Other
Inanimate (<i>n</i> =91)	11%	9%	4% 24%	40%	36% 76%
Animate (<i>n</i> =20)	40%	10%	0% 50%	45%	5% 50%
Human (<i>n</i> =30)	30%	30%	7% 67%	7%	26% 33%

In Kune, the situation is similar although a hierarchy is less pronounced as shown in Table 7. Inanimates in Kune are slightly more likely to be predicated of a location via a stance verb than in Kukatja (being used 33% of the time rather than 23%). However, posture verb strategies do not become quite as pervasive as one moves up the animacy hierarchy: posture verbs only being observed 52% of the time compared to 66% in Kukatja. However, the Kune picture changes depending on how one counts the positional verb *-werndi* 'be high', which involves the posture verb element *-di* 'stand'. If tokens of *-werndi* are added to the core set of posture verbs in Kune, the overall use of elements with postural semantics goes up significantly, counting for 44%, 56% and 63% for inanimate, animate and human figures respectively. If we take this slightly more permissive set of what

constitutes a ‘posture verb’, then we observe two findings: the differences in predicational strategy for humans and animates becomes rather minor in Kune and the use of a ‘posture verb’ strategy for inanimates is much higher in Kune than in Kukatja.

Table 7: Kune strategies for predicating location in DELOCA and TPRS

Figure type	Posture verb			Positional verb	Non-posture verb	
	<i>ni-</i> ‘sit’	<i>di-</i> ‘stand’	<i>yo-</i> ‘lie’	<i>wern.di</i> ‘be.high’	Verbless	Other
Inanimate (<i>n</i> =539)	2%	23%	8%	11%	11%	45%
			33%	11%		56%
Animate (<i>n</i> =121)	22%	21%	8%	15%	8%	36%
			41%	15%		44%
Human (<i>n</i> =115)	21%	13%	18%	11%	1%	36%
			52%	11%		37%

The findings for Wik-Mungkan (shown in Table 8) pattern in some ways like Kukatja on the one hand and like Kune on the other. Like Kukatja, posture verbs are a relatively disfavoured strategy for predicating location of an inanimate figure (just 32% of scenes were described in this way). Like Kune and unlike Kukatja, however, there was no major difference in predicational strategies for scenes involving human and animate figures—both utilising posture verbs for over 60% of the scenes described.

Table 8: Wik-Mungkan strategies for predicating location in DELOCA and TPRS

Figure type	Posture verb			Non posture verb	
	<i>nyiin</i> ‘sit’	<i>than</i> ‘stand’	<i>wun</i> ‘lie’	Verbless	Other
Inanimate (<i>n</i> =258)	2%	9%	22%	45%	22%
			33%		67%
Animate (<i>n</i> =60)	33%	2%	27%	28%	10%
			62%		38%
Human (<i>n</i> =55)	31%	15%	20%	16%	18%
			66%		44%

The data thus support a view that there are differing degrees of grammaticalisation of posture verbs across the three languages when one compares the relative frequency of individual posture verbs with non-human figures. For inanimates, there is evidence of a single posture verb being preferred in all three languages. In Kukatja, the 'sit' member was the most frequent choice and the 'lie' member was barely used. In contrast, for Kune, it is the 'stand' member *-di* (and the related compound *-werndi* 'be elevated') that is the preferred posture verb (43%) while *-ni* 'sit' is only minimally attested (being used for 2% of the relevant scenes). Finally in Wik-Mungkan it is *wun-* 'lie' that is preferred (22%) followed distantly by *than-* 'stand' and *nyiin-* 'sit' (2%). Thus for all languages, the counts of posture verbs used with inanimate figures reflected the qualitative impression that there is a 'default' verb for inanimate figures.

For non-human animates, the three languages showed some un-anticipated results, however. For Wik-Mungkan speakers, the location of non-human animates was rarely described with locative clauses involving *than-* 'stand', which tends to be reserved for either humans or profiling the upright configuration of certain inanimate entities. For scenes where 'stand' might have been anticipated (e.g. a dog 'standing' on all fours), verbless strategies were instead observed. For Kukatja, *ngari-* 'stand' is used with animate non-humans but *ngarri-* 'lie' was entirely unattested for such figure types. Kune speakers utilised all three posture verbs with animate non-humans but, like Kukatja speakers, dispreferred the use of the 'lie' verb.¹³

3.5 Non-posture verb strategies

In the preceding sections, it was shown that a significantly large number of descriptions emerged utilising a fully verbless predication strategy for locative descriptions. However, discussion of alternatives with various consultants and running the task with multiple speakers made it clear that supplying a posture verb was nearly always possible in all languages, reflecting a flexibility in strategy choice.

Other construction types used in the surveyed languages to express locative-existential predication are sampled below. In all three languages, a resultative

¹³Since the stimuli utilised in this quantitative study were not specifically controlled for equal numbers of scenes depicting each anticipated posture type, we did not anticipate equal proportions of each individual posture verb to be used. However, the differences observed here go well beyond variation in scenes depicted. It is an area of future study to carefully investigate posture verb use while controlling carefully for scene type (beyond referential properties of the figure).

or impersonal construction is attested for a number of locational scenes, particularly those depicting small manipulable inanimates objects as figures (cf. Levinson & Wilkins 2006b: 50) (50). For some scenes, Kune speakers make use of a ‘have’ predicate (51), while Kukatja speakers can make use of proprietive case marking on the figure for certain figure-ground relationships (52), especially those where the figure is perceived as a characteristic property of the ground relation.

- (50) Kune: resultative (Bangardidjan Tara Rostron; KLLC_20220805b)
*Nane bid-no **birri-kurrme-ng** durn-no darrme.*
DEM finger-PRT 3PL>3SG-put-PST string-PRT LOC
‘This peg (lit. ‘finger’) is on the line’ (lit. “they” put this peg on the line’)
- (51) Kune: ‘have’ construction (Bulanj Simon Brian; KLLC_20210917a)
*Hill ka-**karrme** man-me.*
hill 3MIN-have.NPST III-food
‘There is food on the hill.’ (lit. ‘the hill “has” food’)
- (52) Kukatja: proprietive case (George Lee: LM2019-04-15-150744)
*Tjurnu nyi-nama **kalyu-kurlu.***
rockhole sit-IPFV water-PROP
‘There is water in the rockhole.’ (lit. ‘the rockhole was “with” water’)

4 Conclusion

We have examined novel and existing material on three Australian Aboriginal languages (Kukatja, Kune, and Wik-Mungkan). The findings for Kukatja, Kune, and Wik-Mungkan generally align with the previous documentation of Australian language posture verbs as set out in §1.2, but careful semantic analysis and quantification reveal underappreciated variation and diversity.

In §3.1 we investigated the lexical semantic space covered by the three posture verbs in each language. Beyond their core stance functions (‘sit’, ‘stand’, ‘lie’), they exhibit overlapping but non-uniform extensions into various other senses such as ‘sleep’, ‘reside’, or ‘wait (for)’. For example, the sense ‘reside’ in Kukatja and Kune aligns with the ‘sit’ verb, whereas in Wik-Mungkan the meaning is associated with *wun-* ‘lie’. Some of these sense types are also characterised by different argument structures for the verb (e.g. *nyina-* ‘sit’ and *ngari-* ‘stand’ in Kukatja take a dative non-subject argument in their sense ‘wait’). We also illustrated how a posture verb can be used for sense selection for polysemous figure

nouns. Our detailed semantic analysis of the terms enriches existing descriptions of posture verbs typologically (Newman 2002a, 2022).

In §3.2 we examined how posture verbs feature in strategies for the predication of location and existence, highlighting that for all three languages it is not always possible to analytically separate the postural meaning from a copula-type use. Overall, it is only when the semantics of a posture verb are incongruent with the arrangement of the figure that it is possible to make an unambiguous determination of a copula use of the posture verb. We also provided evidence that posture verb use in locative and existential predication in all three languages is generally flexible rather than grammatically obligatory.

In §3.3 we showed that for Kukatja and Kune, there is no clear distinction between locative and existential clause types, whereas Wik-Mungkan distinguishes the two based on obligatory encoding of definiteness of the figure in the former.

Finally in §3.4, we quantified some of the variation in posture verb use identified in §3.2 and demonstrated that each of the languages differs with respect to their overall use of posture verbs in locative-existential predication. An animacy hierarchy was shown to be clearly operative in Kukatja and influential in Kune and Wik-Mungkan—higher order animates being more likely to be predicated of a location with an overt posture verb than inanimates. Frequency usage was shown to support our impressionistic observations that just one posture verb in each language was favoured for predicating the location of inanimate figures: *nyina*- ‘sit’ for Kukatja; *-di* ‘stand’ for Kune and *wun*- ‘lie’ for Wik-Mungkan.

The present study suggests that there is some benefit to reviewing existing claims of both posture verbs and copula predication in Australian Aboriginal languages in conjunction with quantified standardised tasks or corpus data. This would pave the way for a renewed and more informed typology of copula predication on the continent, especially given the number of detailed grammatical descriptions published since Dixon’s (2002b) study.

Abbreviations

ACT	action verbalizer	I	noun class I
AUG	augmented number	IDENT	identificational
AVERT	avertive	II	noun class II
CIS.LOC	cislocative (“hither”)	III	noun class III
CONJ	conjunction	INCH	inchoative
D	dative (bound pronoun)	IV	noun class IV
EP	epenthetic	L	locational (bound pronoun)

MED	medial	POT	potential
MIN	minimal number	PROP	propriative
NPP	non-past progressive	PRT	part noun marker
NPST	non-past	RDP	reduplication
NS	non-subject	SEMBL	semblative
OPT	optative	UA	unit augmented
PERL	perlative		

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